

tf.compat.v1.metrics.mean Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/metrics/mean

Computes the (weighted) mean of the given values.

Function Signatures

```
tf.compat.v1.metrics.mean( values, weights=None,  
metrics_collections=None, updates_collections=None, name=None  
)
```

Code Examples

```
tf.compat.v1.metrics.mean(  
values,  
weights=None,  
metrics_collections=None,  
updates_collections=None,  
name=None  
)
```

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tf.compat.v1.metrics.mean(  
values,  
weights=None,  
metrics_collections=None,  
updates_collections=None,  
name=None  
)
```

```
mean, update_op = tf.compat.v1.metrics.mean(  
values=values,  
weights=weights,  
metrics_collections=metrics_collections,  
update_collections=update_collections,  
name=name)
```

```
mean, update_op = tf.compat.v1.metrics.mean(  
values=values,  
weights=weights,  
metrics_collections=metrics_collections,  
update_collections=update_collections,  
name=name)
```

```
m = tf.keras.metrics.Mean(  
name=name)  
m.update_state(  
values=values,  
sample_weight=weights)  
mean = m.result()
```

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m = tf.keras.metrics.Mean(  
name=name)  
m.update_state(  
values=values,  
sample_weight=weights)  
mean = m.result()
```

```
g = tf.Graph()
with g.as_default():
    values = [1, 2, 3]
    mean, update_op = tf.compat.v1.metrics.mean(values)
    global_init = tf.compat.v1.global_variables_initializer()
    local_init = tf.compat.v1.local_variables_initializer()
    sess = tf.compat.v1.Session(graph=g)
    sess.run([global_init, local_init])
    sess.run(update_op)
    sess.run(mean)
2.0
```

```
g = tf.Graph()
```

Documentation

Computes the (weighted) mean of the given values.

Migrate to TF2

`tf.compat.v1.metrics.mean` is not compatible with eager execution or `tf.function`.

Please use `tf.keras.metrics.Mean` instead for TF2 migration. After instantiating a `tf.keras.metrics.Mean` object, you can first call the `update_state()` method to record the new values, and then call the `result()` method to get the mean eagerly. You can also attach it to a Keras model with the `add_metric` method. Please refer to the migration guide for more details.

Structural Mapping to TF2

Before:

After:

How to Map Arguments

Before & After Usage Example

Before:

After:

Description

The mean function creates two local variables, total and count that are used to compute the average of values. This average is ultimately returned as mean which is an idempotent operation that simply divides total by count.

For estimation of the metric over a stream of data, the function creates an `update_op` operation that updates these variables and returns the mean. `update_op` increments total with the reduced sum of the product of values and weights, and it increments count with the reduced sum of weights.

If weights is None, weights default to 1. Use weights of 0 to mask values.

Returns

Raises